

Summary of Backpropagation-Free Training

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April 14, 2025

1 Accuracy of Split Learning

I successfully addressed the issue of low accuracy associated with the BP-free method in split learning scenarios when the data is independently and identically distributed (IID). The accuracy achieved by different methods are presented in Table 1.

Upon analyzing the results of last week, I identified that the consistently lower performance of the BP-free method compared to the backpropagation method was due to the activation of a learning rate adaptive adjustment mechanism. Specifically, this mechanism frequently reduced the learning rate to an excessively low level in the BP-free method, hindering the model's ability to learn effectively. Through experimentation, I observed that the final learning rate of the BP-free method decreases as the amount of data available to each client decreases.

Table 1: Accuracy of split learning. (Number of clients = 10)

Training Strategy	Model	Accuracy(%)
Back-Propagation	ResNet 5	70.94
	VGG 8	76.84
BP-Free	ResNet 5	68.28
	VGG 8	73.27

2 Combine BP with BP-Free

In this week's experiment, I explored combining the backpropagation (BP) method with the BP-free method in a split learning framework. Specifically, I employed the BP-free method to compute gradients on the client side and the BP method to compute gradients on the server side. The experimental setup involved configuring the number of users to 1 and varying the number of model layers per client between 1 and 2. I am still trying to optimize it further, the tentative results of these experiments are summarized in Table 2.

At this stage, the combination of the BP and BP-free methods has not yielded significant improvements in performance. I think this inefficiency comes from the optimizer's configuration. During each training round, the optimizer adjusts hyperparameters based on the model's current training status. However, the adjustment direction determined by the optimizer is not strictly aligned with the gradient directions computed by BP. In this combining method, two distinct gradients are used to update separate layers independently: some are computed using the BP-free method and others using the BP method. This discrepancy may create conflicts for the optimizer as it struggles to update the model parameters.

Table 2: Accuracy of split learning for different methods.

Non-IID Setting	Model	Accuracy(%)
All layers use BP	ResNet 5	70.94
	VGG 8	76.84
All layers use BP-Free	ResNet 5	68.28
	VGG 8	73.27
BP-Free=1	ResNet 5	25.51
	VGG 8	33.46
BP-Free=2	ResNet 5	23.17
	VGG 8	29.66

References